METRIC
MIL-PRF-85045/38
6 August 2024

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, FORTY-EIGHT-FIBER BUNDLE, BLOWN OPTICAL FIBER, TWELVE-FIBER ROLLED FLEXIBLE RIBBON SUBUNITS OR TWELVE-FIBER LOOSE TUBE SUBUNITS, CABLE CONFIGURATION TYPES 3 AND 4 (CABLE BUNDLE AND RIBBON CABLE), APPLICATION B (SHIPBOARD), CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

SCOPE:

This specification sheet addresses the requirements of a 48-fiber bundle for blown optical fiber. Coverage includes fiber cable configuration types 3 and 4 and fiber cable classes SM and MM.

Design constraints. Figure 1 shows the fiber bundle construction and provides essential dimensioning detail for a fiber bundle with a 3.1-millimeter (mm) outer diameter that contains 48 fibers, which could be either intermittently held together as rolled flexible ribbon in four 12-fiber subunits or as 48 loose fibers. Fiber bundle construction consists of an overall bundle jacket (with a 3.1-mm diameter) followed by the either loose or ribbonized fiber. Following figure 1 are requirements for design and construction, material, and performance.

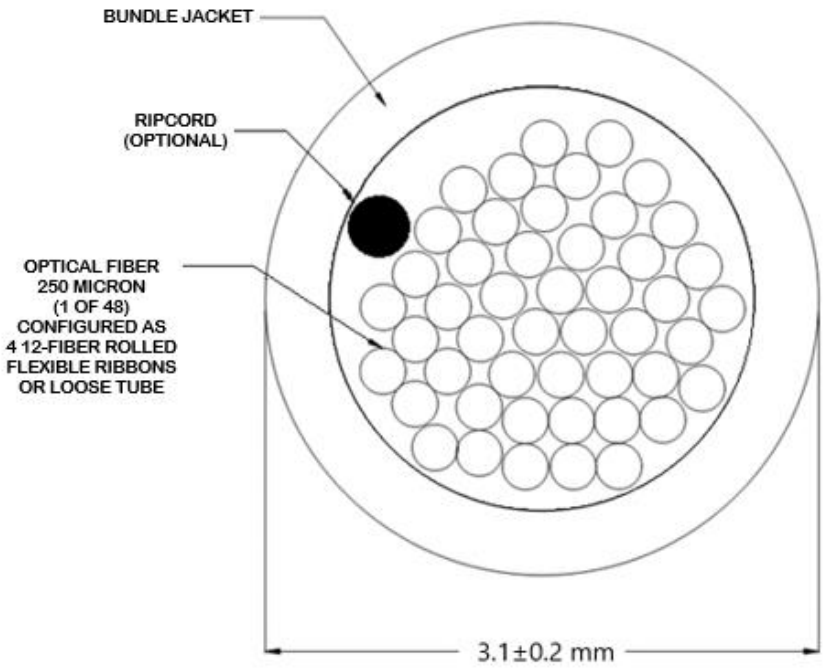


FIGURE 1. Fiber bundle, blown optical fiber, 48-fiber rolled flexible ribbon or loose tube.

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CLASSIFICATION:

Fiber optic cable configuration types: 3 (cable bundle) or 4 (ribbon cable).

Fiber optic cable class: MM (graded-index, glass core and glass cladding, multimode).

SM (dispersion-unshifted, glass core and glass cladding, single-mode).

Part or identifying number (PIN):

48-fiber rolled flexible ribbon fiber bundle (multimode): M85045/38-01R

48-fiber loose tube fiber bundle (multimode): M85045/38-01L

48-fiber rolled flexible ribbon fiber bundle (single-mode): M85045/38-02R

48-fiber loose tube fiber bundle (single-mode): M85045/38-02L

DESIGN AND CONSTRUCTION:

Temperature ranges. Temperature ranges shall be in accordance with table I.

TABLE I. Temperature ranges.

Operating temperature	Non-operating temperature	Storage temperature
-28 to 65 °C (-18 to 149 °F)	-40 to 70 °C (-40 to 158 °F)	-40 to 70 °C (-40 to 158 °F)

Fiber: Class MM fibers shall be in accordance with MIL-PRF-49291/6 (PIN: M49291/6-03).

Class SM fibers shall be in accordance with MIL-PRF-49291/7 (PIN: M49291/7-01).

Buffer diameter. 250±15 micrometers (µm) after coloring (inking).

Fiber color coding. Each optical fiber in a ribbon or loose shall be color coded for identification in accordance with TIA-598 (see 3.4.2.4.1 of MIL-PRF-85045). A visual inspection performed after the temperature life test shall show that the colorant on each optical fiber remains intact with no flaking or other degradation in addition to no change outside specified limits for appearance (color) on the three samples of rolled flexible ribbon.

Rolled flexible ribbon (applicable to M85045/38-01R and M85045/38-02R only):

Configuration. The rolled flexible ribbon shall be configured as an adhesive matrix (bonded layer) applied over 12 adjacent optical fibers that are positioned laterally (in a linear array) with the adhesive matrix only being present intermittently after final manufacturing (see “DEFINITIONS”).

Fabrication. The rolled flexible ribbon shall have no crossovers, defective fibers, or splices.

Ribbon identification marking. Each ribbon (subunit) shall be uniquely identified by marking (color, markings, etc.) for identification in accordance with TIA-598 (see 3.4.2.4.1 of MIL-PRF-85045). A visual inspection performed after the temperature life test shall show that the marking on each subunit remains intact with no flaking or other degradation outside the specified limits for appearance (color/legibility).

Tests for rolled flexible ribbon. Each test sample shall consist of 2 meters of fiber bundle (finished cable) or as specified in the test method. First, the specified testing shall be performed on finished cable; then, the finished cable shall be deconstructed, and the group VII post-test examinations shall be performed on the rolled flexible ribbon lengths. Three samples shall be used for each test or as specified in the test method. Small end caps may be placed on the ends of the samples to eliminate end effects. A plastic cap or epoxy may be used. Test sample length and configuration for the ribbon separation shall be as specified in those tests.

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Ribbon strippability:

Method. One end of each rolled flexible ribbon shall be stripped to a minimum length of 30 mm with commercially available stripping tools to remove the binder (matrix) material and to remove the fiber coating (250-micron buffer). The test shall be performed on both unaged samples (from ribbon dimension test) and specimens of the rolled flexible ribbon from the finished cable that underwent temperature life testing followed by an inspection for ribbon delamination.

Requirement. No fiber breakage in the rolled flexible ribbon shall occur, and any residue of coating that remains shall be readily removed with isopropyl alcohol wipes.

Ribbon delamination:

Method. Applicable except that three separate 2-meter samples of finished cable shall be included in the temperature life test versus performing a test under different conditions. After completion of the temperature test, the test samples of the rolled flexible ribbon (once the outer cable jacket is removed) shall be inspected for ribbon delamination.

Requirement. A visual inspection performed after the temperature life test shall show no delamination occurred on the three samples of rolled flexible ribbon.

Ribbon separation:

Method. The optical fibers in a minimum of 0.5-meter length of rolled flexible ribbon shall be separated either by hand using no tools or by hand using tools without the aid of applied heat.

Requirement. The minimum 0.5-meter length of optical fibers in a rolled flexible ribbon shall be separated with no damage, removal, discoloration, or chemical attack to the 250-micron buffer. Minor removal of buffer colorant is acceptable.

Finished cable:

Dimensions and configuration. See figure 1. An outer jacket shall be applied over 48 optical fibers and a ripcord or filler.

Mass per unit length. ≤6 kilograms per kilometer (kg/km).

Outer cable jacket color:

Class MM fiber: Slate or blue.

Class SM fiber: Yellow.

Jacket material. The overall jacket shall be composed of a low-halogen, low-toxicity polymer material.

Concentricity. Applicable with the following modification. The concentricity shall be measured by first locating the areas where the binder/inner jacket is nearest the jacket outside diameter. From these measurements, the minimum and the maximum wall thicknesses will be selected and used to calculate concentricity. The ratio of the minimum wall thickness to the maximum wall thickness shall be ≥0.50.

Short-term minimum bend diameter. 48 mm. (The short-term minimum bend diameter is to be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter. 48 mm.

Minimum continuous length. The minimum continuous length of all finished cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

Marking. Marking of blown optical fiber bundles is not required. Shipping containers and bundle reels shall be marked.

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PERFORMANCE REQUIREMENTS:

The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet, including the requirements identified in table II.

Optical properties:

Maximum attenuation rate:

3.5 dB per km at 850±20 nanometers (nm) and 1.0 dB per km at 1,300±20 nm for class MM fiber.

0.40 dB per km at 1,310±20 nm and 0.30 dB per km at 1,550±20 nm for class SM fiber.

Bandwidth. Not applicable.

Point discontinuity (attenuation uniformity):

Method. The attenuation uniformity of each individual fiber shall be measured in accordance with TIA-455-78. The attenuation uniformity shall be measured at wavelengths of 850 nm and 1,300 nm for multimode and at 1,310 nm and 1,550 nm for single-mode.

Requirement. There shall be no discontinuities in attenuation along the length greater than 0.1 dB for single-mode and no discontinuities in attenuation along the length greater than 0.2 dB for multimode.

Change in optical transmittance. Measurements shall be made at 1,300±20 nm for multimode and 1,310±20 nm and 1,550±20 nm for single-mode.

Measured fiber positions within the cable. Unless otherwise specified herein, the minimum number of active fibers required to be measured for a change in optical transmittance during mechanical and environmental testing shall be 12.

For shock and vibration testing only, four fiber specimens shall be required to be monitored (one fiber in each subunit/12-fiber grouping). For rolled ribbon only (M85045/38-01R and M85045/38-02R), the fiber specimen chosen from each 12-fiber grouping shall be either fiber 1 (blue) or 12 (aqua). For single-mode fiber shock and vibration testing only, measurements at 1,310±20 nm are required at a minimum (1,550±20 nm is not required).

Crosstalk. Applicable.

Mechanical properties:

Low temperature flexibility. Not applicable.

Cyclic flexing. 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Minimum load shall be 1.0 kg. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable element removability. Applicable with the following additions:

a. The end of the rolled flexible ribbon shall be easily formed by hand into a flat ribbon that fits into an interfacing rectangular trough (such as tool [fusion splice holder], housing [splice tray], and component [MT terminus/connector]).
b. The 12 colored fibers shall be visually inspected to verify that they are retained in the specified order of color once flattened and that 20 centimeters (cm) of matrix material is removed, by either mechanical or chemical methods, from the end of the 12 fibers (except that any residual material left on the fibers from the matrix material shall be removed with a dry cotton cloth).
c. The type of adhesive or matrix material used for the rolled flexible ribbon shall ensure fiber separation does not cause coating damage or leave residue.

Kink resistance (optical fiber cable component [OFCC] kink). Not applicable.

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Hosing. Not applicable.

Dripping. Applicable.

Water absorption. Applicable.

Cable jacket tear strength. Applicable, except the cable jacket tear strength shall be not less than 5 newtons per cm of jacket thickness. Five strips of jacket material shall be used. The test shall be performed in shear orientation on molded samples of the same jacket material: 101.6 mm (4 inches) (long) by 12.7 mm (0.5 inch) (wide) by 3.175 mm (0.125 inch) (thick), with a single longitudinal slit along the length of the specimen.

Cable shrinkage. Applicable.

Wicking. Not applicable.

Environmental properties:

Temperature cycling. The cable shall be tested in accordance with TIA-455-3 using the setup and procedure specified in measurement 3301 of MIL-STD-1678-3 with the temperature profile of "General applications". The cable shall be visually examined after the test. Test fixtures, if used, shall be of minimum mass and approved by the qualifying activity. No other mass shall be added inside the chamber. The change in optical transmittance shall be measured during and after the test. At a minimum for the measurements made during testing, an optical transmittance measurement shall be performed toward the end of each soak period (maintain step) after every cycle. The cable shall be visually examined after the test.

Temperature humidity cycling. Applicable except that the change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Storage temperature. Applicable.

Temperature life (life aging). Applicable. A visual inspection performed after the temperature life test shall include inspection for the requirements specified under ribbon delamination and colorant and for ribbon identification marking.

Thermal shock. Applicable.

Weathering. Not applicable.

Fluid immersion. Not applicable.

Vibration. Not applicable.

Shock. Applicable. The fiber bundle shall be tested within a tube cable. Optical discontinuities shall be measured during the test. A minimum of four optical specimens (fibers), one fiber from the four 12-fiber groupings (selected optical specimen shall be from position 1 [blue] or 12 [aqua] from each grouping), shall be monitored for discontinuity. The change in optical transmittance shall be monitored after the test. The specimens shall be visually examined after the test. The requirement for optical discontinuity shall be met during the test, and the requirement for the change in optical transmittance shall be met after the test.

Flame extinguishing and smoke generation. Applicable. Tube ends shall be plugged with a non-flammable sealant to simulate end caps. Testing shall be performed in the following configuration: a cable in accordance with MIL-PRF-85045/26 that contains one 48 fiber bundle.

Paint susceptibility. Not applicable.

Chemical properties:

Acid gas generation. <5.0 percent.

Halogen content. <0.2 percent.

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Toxicity index. Applicable.

Qualification and conformance inspection. See table II. This table specifies applicable requirements, qualification inspection, and conformance inspection per this specification sheet. The “Requirement paragraph” and “Test paragraph” columns within this table reference MIL-PRF-85045 paragraphs. The “Cable length” column applies only to the qualification inspection. A reduced sample set is allowed for conformance inspection; refer to MIL-PRF-85045 for conformance inspection sample size.

TABLE II. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each 5/	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each 6/	A
	Point discontinuity	7/	7/	Three samples, 0.5 km each 6/	--
II	Crosstalk	3.5.3	4.7.4.3	Three samples, 0.5 km each 6/	--
III	Temperature cycling	3.7.1	7/	Two samples, 0.5 km each 6/ (one on reel, one off)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each 8/	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each 8/	--
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each 9/ (two specimens at each temp)	--
	Temperature life (life aging)	7/	7/	Two specimens, 300 m each 10/	C
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each 10/	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each 10/	C
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.5 km each 6/ (on reel)	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m 11/	--
	Shock	7/	7/	One specimen, 30 m 11/	--
V	Dripping	3.6.13	4.7.5.13	One sample, 30 cm 11/	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips 12/	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m each 11/	C
	Flame extinguishing and smoke generation	3.7.12.2	4.7.6.12.2	One specimen, 50 m 11/	C
	Water absorption	3.7.14	4.7.6.14	Two specimens, extruded jacket material strips 12/	C
VI	Acid gas generation	3.8.1	4.8.1	One specimen, 1 m 13/	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m 13/	C
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m 13/	C

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TABLE II. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/, 3/, 4/	Conformance inspection
VI	Smoke index	3.8.5	4.8.6	One specimen, 1 m 13/	C
VII 14/	Ribbon strippability	7/	7/	Three specimens, 2 m each 14/	--
	Ribbon delamination	7/	7/	Three specimens, 2 m each 14/	--
	Ribbon separation	7/	7/	One specimen, 0.5 m 14/	--

NOTES:

1/ Tolerance on 0.5-km length shall be ±5 percent, provided that results are normalized to 1 km.

2/ Tolerance on shorter lengths shall be ±5 percent.

3/ A sample shall be the initial 0.5-km cable length.

4/ A specimen shall be a cable length that is cut from the initial 0.5-km cable length (i.e., the sample) or is cut from a longer previous cut length (i.e., cut from a longer specimen).

5/ The visual and mechanical inspection shall only be conducted on a 2-meter (m) section of each sample.

6/ The same samples as used in the visual and mechanical inspection shall be used.

7/ As stated under the applicable tests in this specification sheet.

8/ The same samples as used in the temperature cycling test shall be used.

9/ Three specimens cut from each sample used in the temperature cycling test shall be used.

10/ A specimen cut from each sample used in the temperature cycling test shall be used.

11/ A specimen cut from the specimen used in the thermal shock test shall be used.

12/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition of the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 of MIL-PRF-85045 and obtained from flat extruded material that underwent the thermal shock test shall be used.

13/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

14/ Group VII tests are only applicable to M85045/38-01R and M85045/38-02R PINs.

Qualification and conformance inspection by similarity:

Similarity for single-mode and multimode. Manufacturers who are qualified under this specification sheet for the rolled flexible ribbon fiber bundle or loose tube fiber bundle with single-mode fiber (M85045/38-02R or loose tube fiber bundle M85045/38-02L) and whose rolled flexible ribbon fiber bundle or loose tube fiber bundle with multimode fiber passes the tests and inspections specified in table III as specified herein shall be qualified under this specification sheet for the rolled flexible ribbon fiber bundle or loose tube fiber bundle respectively with multimode fiber (M85045/38-01R or M85045/38-01L). This qualification by similarity shall be applicable if the only difference between the previously qualified fiber bundle and the fiber bundle under test is that the optical fiber has been changed from single-mode optical fiber to multimode optical fiber.

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TABLE III. Qualification and conformance by similarity for M85045/38-01R or M85045/38-01L cable (when qualified to M85045/38-02R or M85045/38-02L cable).

Group	Qualification inspection	Qualification inspection M85045/38-01R or M85045/38-01L 1/, 3/	Conformance inspection M85045/38-01R or M85045/38-01L 2/, 3/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	C
	Storage temperature	X	--
	Cyclic flexing	X	--
IV	Thermal shock	X	--

NOTES:

1/ Where manufacturers are qualified for M85045/38-02R or M85045/38-02L and are pursuing qualification for M85045/38-01R or M85045/38-01L, respectively.

2/ Where manufacturers are qualified for M85045/38-01R or M85045/38-01L and M85045/38-02R or M85045/38-02L and are performing conformance testing for M85045/38-02R or M85045/38-02L, respectively.

3/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Similarity for 48-fiber bundle with 72-fiber bundle (rolled flexible ribbon). Manufacturers who are qualified under specification sheet MIL-PRF-85045/39 for the 72-fiber bundle cable (M85045/39-01R or M85045/39-02R) and whose 48-fiber rolled flexible ribbon fiber bundle cable (M85045/38-01R or M85045/38-02R) passes the tests and inspections specified in table IV as specified herein, are qualified under this specification sheet for the 48-fiber bundle cable for the same fiber type (multimode or single-mode). This qualification by similarity shall be applicable only if the rolled flexible ribbon fiber bundle subunit and the bundle cable jacket for the unqualified 48-fiber bundle cable are the same as those for the previously qualified 72-fiber bundle cable. This qualification by similarity shall be applicable if the only difference between the previously qualified cable and the cable under test is that the fiber count has been changed from 72 fibers to 48 fibers. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE IV. Qualification and conformance by similarity for M85045/38-01R or M85045/38-02R rolled flexible ribbon bundle cable (when qualified to M85045/39-01R or M85045/39-02R cable).

Group	Qualification inspection	Qualification inspection M85045/38-01R or M85045/38-02R 1/, 3/	Conformance inspection M85045/38-01R or M85045/38-02R 2/, 3/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Cyclic flexing	X	--
	Temperature humidity cycling	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/39-01R or M85045/39-02R and are pursuing qualification for M85045/38-01R or M85045/38-02R, respectively.

2/ Where manufacturers are qualified for M85045/39-01R or M85045/39-02R and M85045/38-01R or M85045/38-02R and are performing conformance testing for M85045/38-01R or M85045/38-02R, respectively.

3/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Similarity for 48-fiber bundle with 12-fiber bundle (loose tube). Manufacturers who are qualified under specification sheet MIL-PRF-85045/27 for the 12-fiber bundle cable (M85045/27-01L or M85045/27-02L) and whose 48-fiber rolled flexible ribbon fiber bundle cable (M85045/38-01L or M85045/38-02L) passes the tests and inspections specified in table V as specified herein, are qualified under this specification sheet for the 48-fiber bundle cable for the same fiber type (multimode or single-mode). This qualification by similarity shall be applicable only if the loose tube fiber bundle subunit and the bundle cable jacket for the unqualified 48-fiber bundle cable are the same as those for the previously qualified 12-fiber bundle cable. This qualification by similarity shall be applicable if the only difference between the previously qualified cable and the cable under test is that the fiber count has been changed from 12 fibers to 48 fibers. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE V. Qualification and conformance by similarity for M85045/38-01L or M85045/38-02L loose tube bundle cable (when qualified to M85045/27-01L or M85045/27-02L cable).

Group	Qualification inspection	Qualification inspection M85045/38-01L or M85045/38-02L 1/, 3/	Conformance inspection M85045/38-01L or M85045/38-02L 2/, 3/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Cyclic flexing	X	--
	Temperature humidity cycling	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/27-01L or M85045/27-02L and are pursuing qualification for M85045/38-01L or M85045/38-02L, respectively.

2/ Where manufacturers are qualified for M85045/27-01L or M85045/27-02L and M85045/38-01L or M85045/38-02L and are performing conformance testing for M85045/38-01L or M85045/38-02L, respectively.

3/ Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order shall be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

DEFINITIONS:

Flexible. The term flexible is used to emphasize the requirement that the application of the adhesive matrix is applied intermittently to allow the rolled flexible ribbon to take a planar shape for connector termination and splicing and take a circular shape when rolled longitudinally within the cable.

Rolled. The term rolled is used to emphasize the requirement for the ribbon to be rolled longitudinally within the cable to increase the fiber packing density (allow for a specified number of fibers for the given cable diameter).

Referenced documents. In addition to MIL-PRF-85045, this document references the following:

MIL-PRF-49291/6
MIL-PRF-49291/7
MIL-PRF-85045/26
MIL-PRF-85045/27
MIL-PRF-85045/39
MIL-STD-1678-3
TIA-455-3
TIA-455-78
TIA-598

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Review activities:
Army – AR, AV, MI
Navy – AS, EC, OS, YD
Air Force – 02, 13, 19, 70, 93
DLA – CC
MISC – MDA

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
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